

2 Simple Proof

This section presents a simple proof of the Apex Minor Theorem. The next lemma is the key. A *centre* of a graph G is a vertex $\alpha \in V(G)$ such that $\max\{\text{dist}(\alpha, v) : v \in V(G)\}$ is equal to the radius of G .

Lemma 10. *Let A be a planar graph such that $A - z$ is a minor of the $k \times k$ grid for some $z \in V(A)$. Let G be an A -minor-free graph with radius r and centre α . Then $G - \alpha$ has no $k^r \times k^r$ grid minor.*

Proof. We proceed by induction on r . In the $r = 0$ case, $V(G - \alpha) = \emptyset$ and the result holds. Now consider the radius r case, and assume the radius $r - 1$ case holds. Let $V_i := \{x \in V(G) : \text{dist}_G(x, \alpha) = i\}$ for $i \in \{0, \dots, r\}$. So $V_0 = \{\alpha\}$ and $V_0 \cup \dots \cup V_r = V(G)$.

Suppose for the sake of contradiction that $G - \alpha$ has a $k^r \times k^r$ grid minor. Partition this $k^r \times k^r$ grid into $k \times k$ subgrids, so that contracting each subgrid to a vertex gives a $k^{r-1} \times k^{r-1}$ grid.

First suppose that some $k \times k$ subgrid is contained in V_r . Thus H is a minor of $G[V_r]$. By construction, $G[V_0 \cup \dots \cup V_{r-1}]$ is connected. Let G' be obtained from G by contracting $V_0 \cup \dots \cup V_{r-1}$ to a vertex w . Since every vertex in V_r has a neighbour in V_{r-1} , every vertex in G' is at distance at most 1 from w . Since H is a minor of $G[V_r]$, A is a minor of G' and thus of G , which is a contradiction.

Now assume that every $k \times k$ subgrid intersects $V_1 \cup \dots \cup V_{r-1}$. Let G' be obtained from G by contracting each subgrid to a vertex, and deleting any vertices in V_r not in one of the subgrids. So every vertex in G' is at distance at most $r - 1$ from α , and $G' - \alpha$ contains a $k^{r-1} \times k^{r-1}$ grid minor. Since G' is a minor of G , G' is A -minor-free. Hence G' contradicts the assumed truth of the $r - 1$ case.

Therefore, $G - \alpha$ has no $k^r \times k^r$ grid minor. □

Lemmas 2 and 10 imply that for every apex graph A with t vertices, every A -minor-free graph G with radius r has no $(2t - 2)^r \times (2t - 2)^r$ grid minor. Equation (1) implies that $\text{tw}(G) \in O^*((2t)^{9r})$. This completes our simple proof of the Apex Minor Theorem by Eppstein [18] (Theorem 4).

3 Polynomial Upper Bound

This section proves Theorem 5 from Section 1. We need the following straightforward lemma.

Lemma 11. *If a graph G is a minor of the $k \times \ell$ grid, then there is an H -model $(B_u : u \in V(H))$ in the $2k \times 2\ell$ grid J , such that for each vertex $u \in V(H)$ there is a vertex h_u in B_u incident to no edge of J representing an edge of H .*